



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Certified Lean Six Sigma White Belt (CLSSWB) Training

TGS Ref No: TGS-2025053210

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 2

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1	1 July 2026	Initial release — CLSSWB 1-day lesson plan for the Lean Six Sigma awareness course.	Dr. Alfred Ang
2	19 July 2026	Lesson plan rebuilt from the single-source content module. Schedule restructured to follow the DMAIC roadmap end to end within one 8-hour training day, with exactly one hands-on lab per DMAIC phase (5 labs) on a single continuous scenario. Content simplified to White Belt awareness depth: sigma-level and DPMO calculation, MSA, FMEA, value stream mapping, Kano analysis, weighted solution-selection matrices and SPC control limits removed. Assessment aligned to the TMS record: WA (SAQ, 30 min) plus Case Study (30 min); slide references added to the schedule.	Dr. Alfred Ang

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Course Information

Course Title	Certified Lean Six Sigma White Belt (CLSSWB) Training
WSQ Course Reference	TGS-2025053210
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	1 day · 8 training hours
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on Lean Six Sigma labs using the BrewBean Cafe morning rush improvement scenario
TSC Alignment	Quality Process Control (ELE-QUA-5006-1.1)
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Describe the core concepts of quality, Lean, Six Sigma and Lean Six Sigma, and the belt roles in an improvement team.
- LO2: Explain the five phases of the DMAIC roadmap and what happens in each.
- LO3: Identify customer requirements and a clear problem statement in the Define phase.
- LO4: Describe how a process is mapped and measured, and identify the eight wastes in the Measure phase.
- LO5: Identify likely causes of a problem using 5 Whys and Fishbone analysis in the Analyze phase.
- LO6: Describe common improvement and control actions that fix a cause and hold the gain.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 30 minutes, open book.
- Case Study (CS) — applied Lean Six Sigma scenario tasks, 30 minutes, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 2 from 4:00 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — Lean Six Sigma foundations and the full DMAIC roadmap, with assessment

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, ground rules and mandatory

		digital attendance (AM) [slides 2–15]
10:00–11:00	60 min	FOUNDATIONS — What is Quality; What is Lean; What is Six Sigma; Lean vs Six Sigma vs Lean Six Sigma; the belt roles and where a White Belt contributes; the DMAIC roadmap [slides 16–26]
11:00–11:15	15 min	Tea break
11:15–12:00	45 min	DMAIC · DEFINE — Voice of the Customer; VOC to CTQ translation; problem statements; SMART goals; the project charter and scope [slides 27–39]
12:00–12:30	30 min	Hands-on: Lab 1: Define — Customer Requirements and the Problem Statement [slides 40–45]
12:30–13:30	60 min	Lunch break
13:30–14:15	45 min	DMAIC · MEASURE — process mapping; SIPOC; types of data; data collection plans; check sheets; the eight wastes (DOWNTIME); value-added analysis [slides 46–57]
14:15–14:45	30 min	Hands-on: Lab 2: Measure — Map the Process and Spot the Waste [slides 58–64]
14:45–15:30	45 min	DMAIC · ANALYZE — symptom vs root cause; 5 Whys; Fishbone (5M); reading a Pareto chart; common vs special cause variation [slides 65–76]
15:30–15:45	15 min	Tea break
15:45–16:15	30 min	Hands-on: Lab 3: Analyze — Find the Root Cause using the 5 Whys and Fishbone tools [slides 77–83]
16:15–16:35	20 min	DMAIC · IMPROVE — generating solutions; impact/effort screening; 5S; mistake proofing (Poka-Yoke); standard work; piloting [slides 84–94]. Hands-on: Lab 4: Improve — Choose and Pilot a Countermeasure [slides 95–101]
16:35–16:55	20 min	DMAIC · CONTROL — the control plan; visual management; SOPs; team huddles; handover [slides 102–110]. Hands-on: Lab 5:

		Control — Hold the Gain and Hand Over [slides 111–117]
16:55–17:30	35 min	Course recap, revision and Briefing for Assessment [slides 118–128]
17:30–18:00	30 min	Written Assessment (WA) — Short-Answer Questions (SAQ), 30 minutes, open book
18:00–18:30	30 min	Case Study (CS) — applied Lean Six Sigma scenario tasks, 30 minutes, open book. PM digital attendance

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to the DMAIC phases)

DMAIC phase / Topic	Weighting	Labs
FOUNDATIONS: Lean Six Sigma Foundations	20%	
DEFINE: Define — Understand the Problem	20%	Lab 1
MEASURE: Measure — See What Is Really Happening	20%	Lab 2
ANALYZE: Analyze — Find the Cause	20%	Lab 3
IMPROVE: Improve — Fix the Cause	10%	Lab 4
CONTROL: Control — Hold the Gain	10%	Lab 5